Reading Identity from Pixels: A Controlled Comparison of Eigenfaces and Deep Face Embeddings, with a Webcam Case Study on Illumination

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Abstract

Face recognition compares images by turning each into a vector and measuring distances. We compare original pixel vectors (with Eigenfaces PCA for visualisation) to vectors from a deep network pretrained on millions of faces. This paper reports a controlled replication and measurement study of how much the second recipe buys, and where its claimed robustness to illumination comes from. On the 400-image Olivetti Faces dataset we compare a raw-pixel baseline (4096-d; Eigenfaces PCA for plots) with a frozen VGGFace2-pretrained Inception-ResNet-v1 embedding (512-d, unit norm), using identical PCA settings for visualisation and computing every metric in full dimensionality. We add a known-population (closed-set) verification analysis under one protocol for both representations, and a webcam field study of one person under six lighting conditions.

The embedding wins on every identity metric: silhouette 0.106 to 0.497, 1-NN accuracy 94.5% to 100%, pair ROC area 0.881 to 0.9995, and known-population verification EER 8.7% to 0.22% (at the protocol's 0.5% false-reject resolution). At a matched 1% false-reject rate, Map A accepts 48.6% of impostors against 0.044% for Map B. The verification protocol contains only known identities, so it measures cross-identity confusion rather than rejection of strangers. Two methodological findings matter as much. The raw-pixel 2-D plot preserves *more* of its space's pairwise geometry (rank correlation 0.807) than the embedding's (0.517), so the prettier picture belongs to the worse space. In the one-subject observational webcam study, the front-lit room-versus-flash difference lies within the repeat-scan range, while the mean contrast between labelled flash directions is 3.83 times within-scan jitter; pose and auto-exposure were not controlled. PCA plots are fitted transductively, but recognition metrics use the original unprojected vectors. The pretrained model's training corpus is far larger than the test set, so the comparison describes a gap under specific conditions, not the isolated effect of representation learning or a calibrated live threshold.

1 Introduction

A face recognition system is a distance function wearing a decision rule. An image becomes a vector; two vectors are compared; a threshold turns the comparison into "same person" or "different person". Everything follows from the map: if one identity's vectors land close together and different identities land far apart, one threshold can work; if not, no threshold tuning will save it.

There are two canonical maps. Eigenfaces applies PCA to centred face images (Turk and Pentland, 1991): unsupervised, cheap, interpretable. The modern approach trains a deep network on a large corpus with a metric-learning objective—FaceNet's triplet loss (Schroff et al., 2015), VGGFace2's

3.31 million images (Cao et al., 2018), ArcFace's angular margin (Deng et al., 2019)—and uses it as a fixed feature extractor; NIST's FRVT evaluates such systems at scale (Grother et al., 2019). Does the learned embedding separate identity better, and can the difference be shown with a number rather than a nice-looking picture? That question is usually answered either by a benchmark table that hides the mechanism or by two scatter plots that are an opinion. We wanted a comparison small enough to inspect completely and controlled enough that the difference cannot be attributed to plotting choices.

Framing. This is not a novel-method paper; it is a controlled replication and measurement study with a practitioner's case study attached. The comparison is not "learned versus raw" in any clean sense: it is a pretrained embedding trained on millions of faces versus full-dimensional raw pixels from 400 Olivetti images. PCA sees the probe images when fitting the *plots*, but the headline scores use unprojected vectors; that transductive visualisation does not make the measured gap conservative. Upsampling 64×64 to 160×160 invents no detail for Map B, and its training corpus dwarfs the test set. The honest claim is conditional: under these conditions the pretrained embedding separates identity better, and the gap can be measured.

What was done. Three studies share one codebase and one measurement discipline. (i) *Two maps*: Olivetti's 400 greyscale images of 40 people (Samaria and Harter, 1994), processed as raw pixels (Map A) or frozen embeddings (Map B), with identical PCA settings for visualisation and metrics in full dimensionality. (ii) *One known-population (closed-set) verification protocol for both*: five enrol and five probe per identity, five seeded splits, reporting EER, TAR at fixed FAR, and FAR at matched 1% FRR. (iii) *A webcam illumination field study*: twelve scans of one person, ten frames each, under six lighting conditions—a case study of whether "robust to lighting" holds uniformly, and it does not.

Measurement discipline and findings. Two-dimensional PCA maps are for human eyes; a projection can look convincing while discarding the geometry a classifier uses, and our raw-pixel map preserves more of its true pairwise structure than the embedding's even though the embedding is better by every metric. We quantify the loss with a rank correlation between 2-D and full-dimensional pairwise distances and treat any statement not backed by a full-dimensional number as a hypothesis. The embedding wins every identity metric, and the margin grows as the task becomes more like a real decision: silhouette $0.106 \rightarrow 0.497$, 1-NN $94.5\% \rightarrow 100\%$, pair ROC-AUC $0.881 \rightarrow 0.9995$, known-population verification EER $8.7\% \rightarrow 0.22\%$. For this one subject, the observed front-lit room-versus-flash contrast falls within repeat-scan variation while flash-direction labels accompany mean distances 3.83 times within-scan jitter; the capture is not randomised, so these are not isolated causal effects. The cross-domain distance to Olivetti is large and domain-confounded; we report it only as a descriptive distance, not as an impostor distribution and not as a bound on stranger distances.

What is not claimed. No state-of-the-art performance, new embedding, or calibrated security system is claimed: the verification data are small and pre-aligned from a single session; the webcam study has one subject and no stranger; and the verification protocol uses only the 40 known identities, so it does not measure rejection of unknown people.

2 Background

Verification, identification, and Eigenfaces. *Identification* asks which enrolled identity a probe belongs to; *verification* asks whether it matches a claimed identity. Both need a representation ϕ and a distance d; verification compares $d(\phi(x), \phi(y))$ with a threshold τ. In the open-set setting the claimed identity may not be enrolled, so the system must say "unknown"—the setting of face unlock. Our verification experiment is closed-set: all 40 identities are enrolled, so it measures cross-identity confusion rather than rejection of unknown people. Eigenfaces stacks each centred image as a vector in $\mathbb{R}^{4096}$, computes the leading eigenvectors of the covariance, and projects onto the first k (Turk and Pentland, 1991). The components can be reshaped into images, but the objective is variance, not between-identity separation, and on faces the largest variance is usually illumination. On tightly cropped, roughly aligned data such as Olivetti it is nevertheless a strong baseline, and therefore the right comparator.

Deep face representations. DeepFace showed that a deep network trained on millions of faces could approach human verification performance (Taigman et al., 2014). FaceNet's triplet loss optimises the verification property directly: anchor–positive close, anchor–negative far by a margin (Schroff et al., 2015). VGGFace2 scaled the labelled corpus to 3.31 million images of 9,131 identities (Cao et al., 2018); ArcFace replaced the triplet objective with an additive angular margin (Deng et al., 2019); surveys trace the progression (Wang and Deng, 2021). Here the network is a fixed feature extractor: Inception-ResNet-v1 (Szegedy et al., 2017) pretrained on VGGFace2 via `facenet-pytorch` (facenet-pytorch, 2024), frozen, producing an L2-normalised 512-d vector. Freezing isolates what a pretrained representation contains without confounding the comparison with training that 400 images could not support. For unit vectors, cosine similarity is $1 - d_{\text{eucl}}^2/2$.

Detection and illumination. MTCNN is a cascade: a proposal network (P-Net) scans an image pyramid, a refinement network (R-Net) filters and regresses, and an output network (O-Net) refines and predicts five landmarks (Zhang et al., 2016). The `facenet-pytorch` implementation crops the O-Net box to a square and resizes to 160×160; it returns the landmarks but does not warp the patch, so alignment is detector-box alignment. Olivetti avoids the issue: its faces are pre-aligned, which makes it the best case for raw pixels and an easy case for the deep model. Illumination is the canonical nuisance: Extended Yale Face Database B held pose and expression roughly constant while sweeping light, and illumination-cone models showed that images under all lighting directions lie near a low-dimensional convex cone (Georghiades et al., 2001). Changing the *amount* of light scales pixel values roughly uniformly; changing the *direction* produces shading that moves across the face.

Metrics and visualisation. Verification is characterised by false accept rate (FAR), the fraction of impostor trials accepted, and false reject rate (FRR), the fraction of genuine trials rejected. Sweeping the threshold traces an ROC curve of TAR $= 1 -$ FRR against FAR; summary points are the equal error rate (EER), where FAR $=$ FRR (interpolated here from a finite score grid), and TAR at fixed FAR, which is how operational systems are compared because deployments fix the acceptable false-accept rate rather than the threshold (Grother et al., 2019). FERET established the vocabulary of defined gallery/probe splits and impostor trials (Phillips et al., 2000). We also use discriminability d' under an equal-variance Gaussian assumption,

$$d' = \frac{\mu_{\text{imp}} - \mu_{\text{gen}}}{\sqrt{(\sigma_{\text{gen}}^2 + \sigma_{\text{imp}}^2)/2}}. \tag{1}$$

PCA is a projection, and a projection can lie by omission; t-SNE is the usual cautionary example (van der Maaten and Hinton, 2008), but ordinary PCA scatters have a subtler problem: the two directions of largest variance need not be the two that separate identities. We quantify how much of the full-dimensional geometry a 2-D map preserves using the Spearman rank correlation between 2-D and full-dimensional pairwise distances, and never compute a metric on 2-D coordinates.

3 Related work

Eigenfaces (Turk and Pentland, 1991) is the reference appearance-based method; its linear descendants replaced PCA's unsupervised criterion with label-aware objectives, and its failure modes motivated decades of illumination normalisation. Our Map A is deliberately the plain 1991 recipe, because the point is to measure what the raw representation contains. The deep lineage from DeepFace (Taigman et al., 2014) through FaceNet (Schroff et al., 2015), VGGFace2 (Cao et al., 2018), and ArcFace (Deng et al., 2019) is surveyed by Wang and Deng (2021); NIST's FRVT evaluates submitted systems on millions of images at fixed FAR (Grother et al., 2019). On measurement, the visualisation-versus-embedding distinction is well known (van der Maaten and Hinton, 2008), but the observation that the *worse* representation can have the more faithful 2-D plot is a useful cautionary result. Small, fully specified replications are valuable precisely because they are inspectable.

4 Data

Olivetti Faces. Olivetti (Samaria and Harter, 1994) provides 400 greyscale images of 40 people, ten each, at 64×64 pixels in $[0, 1]$. The images are tightly cropped and roughly eye-aligned: across one person's ten photographs the eyes sit in nearly the same place, and what changes is expression, head tilt, and lighting. This is the best case for raw pixels, because the geometric normalisation pixel methods are most sensitive to has already been done; it is also an easy case for a deep model. All Olivetti results should be read as a best-case, optimistic reading for the pixel representation. For the map comparison every image is used once—nothing is trained, so there is no split—and for verification the images are partitioned per identity into five enrolment and five probe images, identically for both maps. All 40 identities are enrolled and there are no unknown-identity probes, so this protocol measures cross-identity confusion rather than open-set rejection of strangers. The dataset-wide pixel range is $[0, 1]$, and the preprocessing of the 400-image network batch matches the reference recipe exactly.

Webcam scans. Twelve scans of the author's face were captured on 6 September 2026 with a laptop webcam, ten consecutive frames each, each frame processed through the same MTCNN and embedding pipeline. The saved artefact per scan is a $(10, 512)$ float32 matrix of unit-norm embeddings; raw frames were not retained. Conditions: room light from the front and left, and phone flash from the front, left, right, and below, two repeat scans per condition. Room light was only available from two directions, so the design is unbalanced by necessity (twelve scans, not sixteen). Frames within a scan are consecutive webcam frames of a moving person, not independent photographs: mean embedding distance grows from 0.153 at lag 1 to 0.257 at lag 9, so the effective sample size per scan is below ten and scan means are the more meaningful unit. The first frame of a scan is also systematically different (mean distance 0.243 to the rest of the scan at frame 0 versus 0.167 at frame 5), consistent with webcam warm-up.

Model. The embedder is Inception-ResNet-v1 pretrained on VGGFace2 (Szegedy et al., 2017; Cao et al., 2018) via `facenet-pytorch` (facenet-pytorch, 2024). The checkpoint is `20180402-114759-`

`vggface2.pt`, loaded in evaluation mode and frozen. It has 23,482,624 embedding parameters; the unused 8,631-class head adds 4,427,703. Outputs are L2-normalised. Nothing is trained or fine-tuned.

5 Method

5.1 System and preprocessing

Figure 1 shows the system. Olivetti images enter a raw-pixel branch (flattened 4096-d; PCA to 2-D for presentation) and a frozen-embedding branch (preprocess, embed, the same PCA settings for presentation); both feed a shared evaluation module that computes every metric in native dimensionality. The live path reuses the embedding branch with MTCNN on CPU and embedding on Apple MPS.

Both paths must produce the same input domain, because the network's standardisation is part of the model (Figure 2). For Olivetti, each 64×64 greyscale image in $[0, 1]$ is replicated to three identical channels, bilinearly resized to 160×160 with antialiasing, multiplied by 255, and standardised by

$$x \mapsto \frac{x - 127.5}{128}. \tag{2}$$

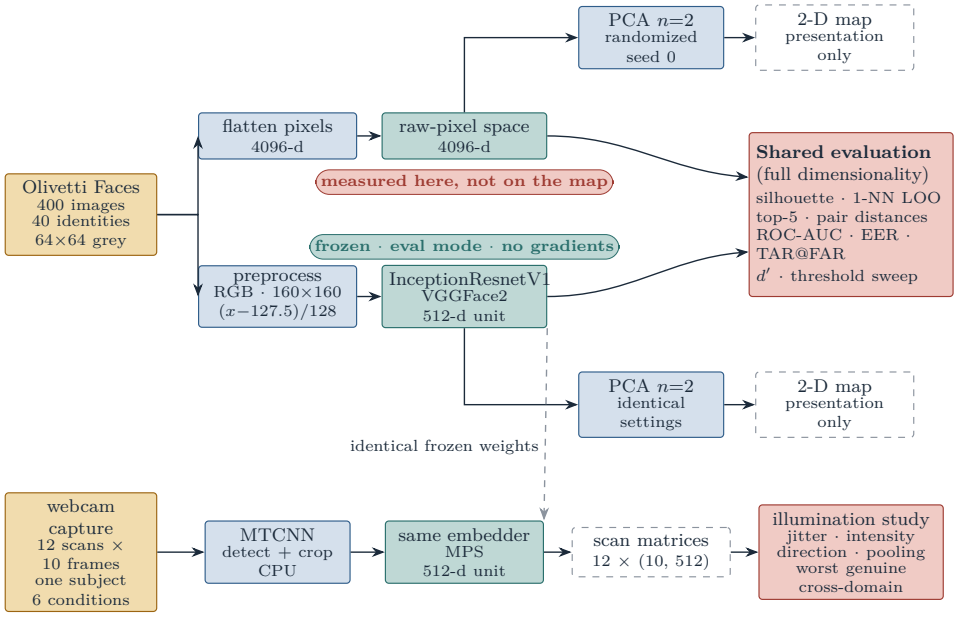
The resize invents no detail, and the recipe reproduces the reference implementation exactly on the 400-image batch. For webcam frames, MTCNN detects a face box with a three-stage cascade, crops and resizes a square to 160×160; the library then converts the crop to a $[0, 1]$ tensor and applies its fixed standardisation $x \mapsto (x \cdot 255 - 127.5)/128$, which is equivalent to $(x - 127.5)/128$ on the original 0–255 intensities. The library crops the O-Net box and does not warp on the five landmarks, so alignment is detector-box alignment; MTCNN runs on CPU because its operations are unsupported on MPS.

5.2 Frozen embedder and PCA maps

Figure 3 shows Inception-ResNet-v1 (Szegedy et al., 2017) with tensor shapes measured by forward hooks on the actual VGGFace2 checkpoint (Cao et al., 2018; facenet-pytorch, 2024). A seven-operation stem is followed by five Inception-ResNet-A blocks (Block35, scale 0.17) at $256 \times 17 \times 17$, reduction A (`mixed_6a`) to $896 \times 8 \times 8$, ten Inception-ResNet-B blocks (Block17, scale 0.10), reduction B (`mixed_7a`) to $1792 \times 3 \times 3$, five Inception-ResNet-C blocks (Block8, scale 0.20) plus a final C block without ReLU, global average pooling to 1792, a bias-free linear layer and batch normalisation to 512, and L2 normalisation. The model is in evaluation mode with gradients disabled, so embedding is deterministic up to hardware arithmetic, and embeddings are cached.

Both 2-D maps use PCA with two components, randomised SVD, seed 0, and sklearn's default centring (Pedregosa et al., 2011); the same settings, colours, and style are used for both. We deliberately do not apply `StandardScaler`: all 4096 pixel columns share one unit, so per-column scaling would inflate near-constant background pixels and change the variance ordering for reasons unrelated to identity; classic Eigenfaces centres but does not standardise. The same implementation produces the explained-variance analysis and the eigenface images; the 2-D coordinates are never a metric input. A rank control projects the raw pixels onto their full rank of 399 components (400 centred samples span at most rank 399), which preserves Euclidean distances exactly (maximum absolute difference 3.5×10^{-12}) and reproduces Map A's metrics to numerical precision. This shows that Map A's nominal 4096-dimensionality is not itself the source of its metrics, but it does not match the two spaces (399 versus 512) or control their intrinsic geometry; this check cannot

Map A — original 4096-d pixels for metrics; PCA is fitted on all images for the 2-D visualisation only.



Map B — pretrained deep embedding; the network never sees Olivetti during training. Arrows show data flow, not training.

Figure 1: End-to-end system. Raw-pixel branch (Map A), frozen-embedding branch (Map B), and the live webcam path. The 2-D PCA maps are for inspection only; all metrics are computed on the full 4096-d and 512-d vectors before projection. The plot PCA is fitted transductively on all 400 images; full-dimensional scores use the unprojected vectors. Map B's frozen network was pretrained on a separate corpus.

determine whether dimensionality contributes to the raw-versus-embedding gap, which is measured rather than mechanistically attributed.

5.3 Metrics and protocols

Distances are Euclidean; for unit vectors, cosine similarity is $1 - d^2/2$. The silhouette score is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad S = \frac{1}{N}\sum_{i=1}^{N} s(i), \tag{3}$$

with $a(i)$ the mean distance to the same identity and $b(i)$ the mean distance to the nearest other identity, on full-dimensional vectors. *1-NN leave-one-out* finds each image's nearest neighbour among the other 399, excluding self; top-5 asks whether the true identity is among the five nearest. *Pair-level* analysis scores all $\binom{400}{2} = 79{,}800$ unordered pairs: 1,800 genuine (40 identities $\times\binom{10}{2}$) and 78,000 impostor. FAR and FRR at threshold τ are

$$\mathrm{FAR}(\tau) = \frac{\#\{d_{\mathrm{imp}} \le \tau\}}{N_{\mathrm{imp}}}, \qquad \mathrm{FRR}(\tau) = \frac{\#\{d_{\mathrm{gen}} > \tau\}}{N_{\mathrm{gen}}}, \tag{4}$$

from which we report ROC-AUC, EER and its threshold, TAR at FAR $= 10^{-3}$ and 10^{-2}, and d' from Equation 1. *2-D fidelity* is the Spearman rank correlation between 2-D and full-dimensional pairwise distances over 20,000 seeded pairs. The *confusion analysis* compares 1-NN predictions

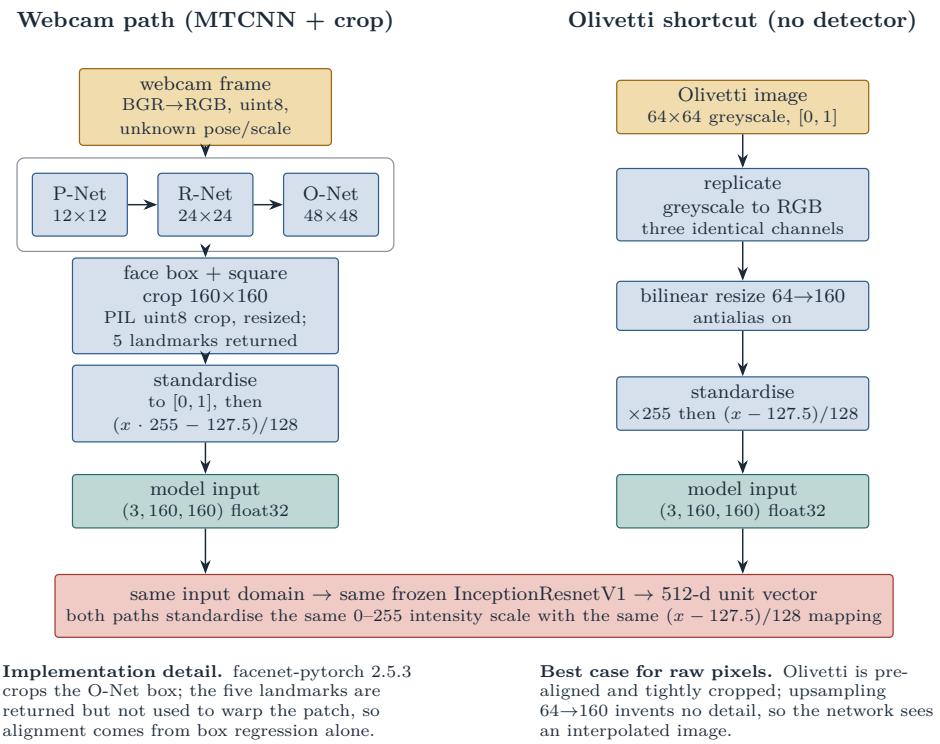
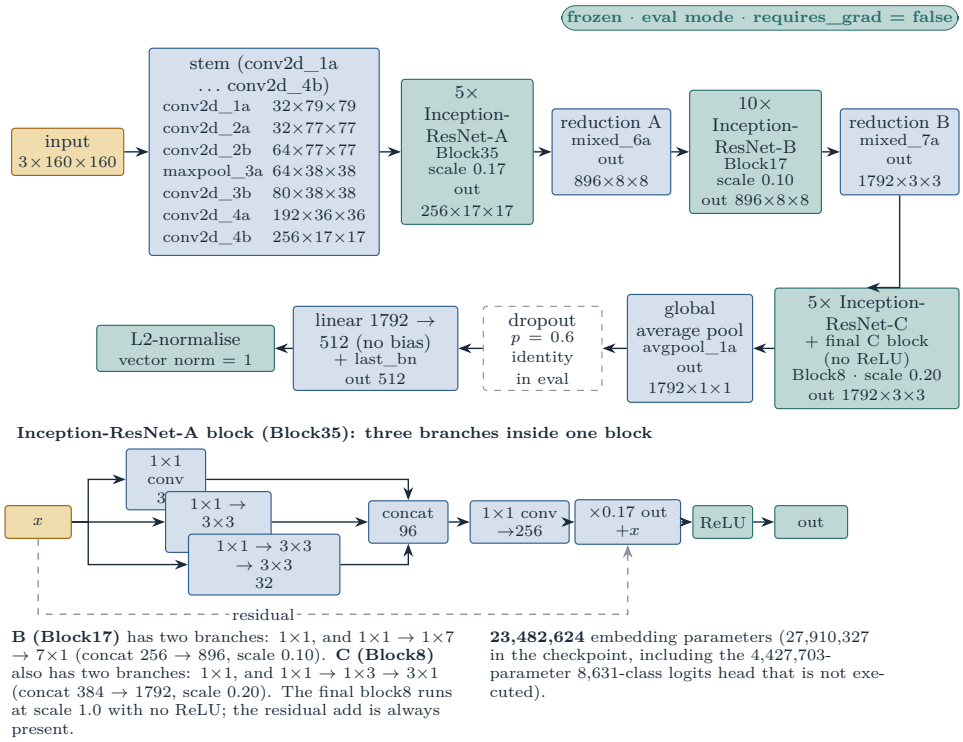


Figure 2: The two preprocessing paths. Webcam (left): MTCNN detects and crops, then the network's fixed standardisation is applied. Olivetti (right): no detector; greyscale is replicated to RGB, resized 64→160 with antialiasing, and standardised identically. The $(x - 127.5)/128$ step is part of the model definition.

across maps; with zero Map B errors the "both wrong" fraction has an empty denominator and is `null`, not zero.

The map comparison uses every image once. Verification uses the known-population (closed-set) protocol in Figure 4: for each of the 40 identities, five images enrol and the other five are held-out probes. Every score is the *minimum* Euclidean distance from a query image to the five enrolment images of the claimed identity, and the decision is accept if the score is at most τ. Genuine trials use each identity's five held-out probes as queries; impostor trials use each of the 390 images of the other 39 identities as queries against the same enrolment set. The primary result averages five random 5/5 splits (seeds 0–4), giving 200 genuine and 15,600 impostor trials per split, identical for both representations. All identities are known and enrolled, so there are no unknown-identity probes and the reported FAR is cross-identity error within a closed population. We report per-split EER, TAR at FAR $= 10^{-3}$ and 10^{-2}, AUC, and FAR at matched 1% FRR, plus a fixed first-half/second-half split (session-bias check) and a mean-template score rule (sensitivity check). These numbers are descriptive, in-sample, known-population cross-identity estimates: every impostor is one of the same 40 enrolled identities, thresholds are chosen and evaluated on the same scores, and EER is interpolated between observed thresholds rather than a realizable operating point. They are not open-set or deployment estimates. An additive identity-disjoint check, with thresholds chosen on calibration identities and evaluated on disjoint and unknown-to-gallery identities, is reported in Section 6.



B (Block17) has two branches: 1×1, and 1×1 → 1×7 → 7×1 (concat 256 → 896, scale 0.10). **C (Block8)** also has two branches: 1×1, and 1×1 → 1×3 → 3×1 (concat 384 → 1792, scale 0.20). The final block8 runs at scale 1.0 with no ReLU; the residual add is always present.

23,482,624 embedding parameters (27,910,327 in the checkpoint, including the 4,427,703-parameter 8,631-class logits head that is not executed).

Figure 3: Inception-ResNet-v1 embedding network. Shapes are measured from `results/model_arch.json` (forward hooks on the VGGFace2 checkpoint, batch dimension dropped) and follow `facenet-pytorch` 2.5.3 naming. The model is frozen: evaluation mode, no gradients, 23,482,624 embedding parameters. The inset shows the Inception-ResNet-A block; B and C have two branches each, with kernel shapes and scales given in the note.

5.4 Live protocol and measurement rule

The twelve scans are analysed at frame and scan level (Figure 5). *Within-scan jitter* is the mean of all 45 frame-pair distances inside a scan: the frame-to-frame noise floor. *Same-condition between-scan* distance compares the repeat scans of each condition. The named *intensity comparison* contrasts room-front and flash-front frames; the named *direction comparison* averages the six flash-direction pairs, and the descriptive ratio divides it by within-scan jitter. Different scans and uncontrolled pose/exposure prevent causal interpretation. *Pooling* compares frame-level distances with distances between re-normalised scan means. The *worst genuine* pair is the largest same-person frame-pair distance, with its cosine equivalent and the same quantity after scan averaging. The *cross-domain* check measures distances from the 400 Olivetti embeddings to the live scans; it is domain-confounded and reported only as a descriptive distance, not as a calibrated impostor distribution and neither as an upper nor a lower bound on webcam-domain stranger distance.

Measure in full dimensionality, plot in 2-D: no metric is computed on PCA coordinates, and the rank correlation quantifies what the plot loses. *One pipeline*: load, embed, project, plot, score, with only the representation swapped. All experiments use seed 0 unless a protocol averages seeds 0–4; reported measurements are manually transcribed from machine-written JSON and require consistency review after regeneration. Hardware and versions are in Section 9.

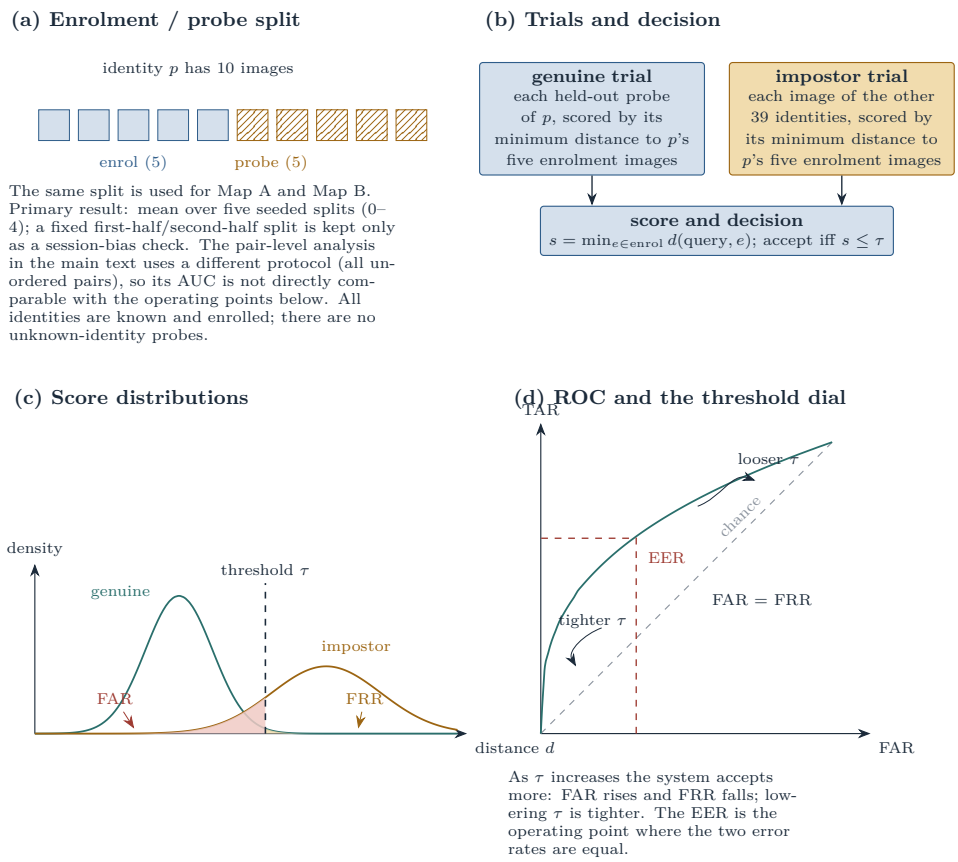


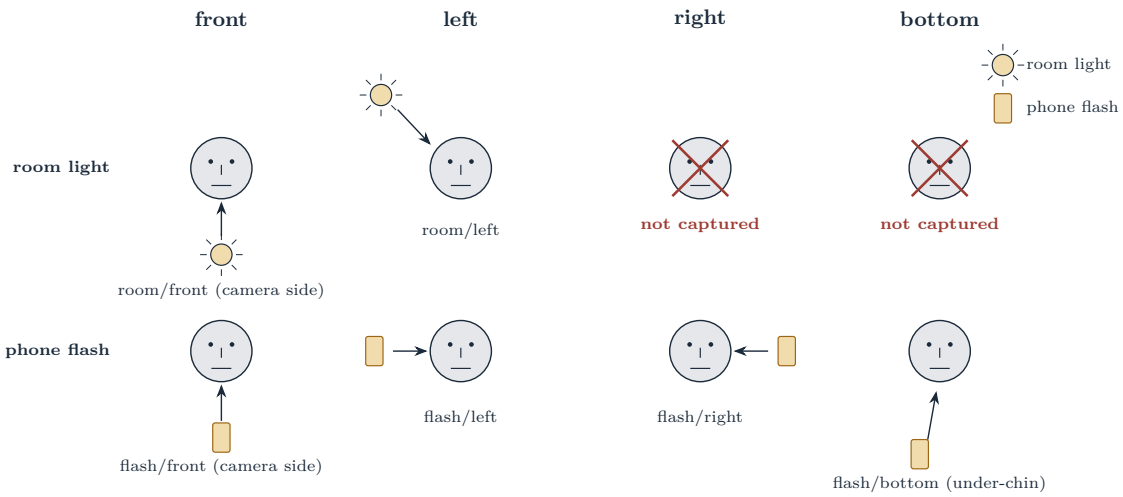
Figure 4: Evaluation protocol. (a) Per identity, five images enrol and five probe; the split is identical for both maps. (b) Genuine and impostor trial definitions. (c) Score distributions with threshold τ; FAR is impostor mass at or below τ, FRR is genuine mass above it. (d) ROC curve and the security–convenience dial: raising the distance threshold τ admits more genuine probes and more impostors, while lowering τ is tighter. Curves in (c) and (d) are schematics, not measured data.

6 Results

6.1 The two maps

Figure 6 shows the two 2-D PCA maps with identical plotting settings. Map A's leading components are brightness gradients, with 23.8% of pixel variance in component 0 and 2, 4, 19, and 66 components for 25%, 50%, 75%, and 90% (95% not reached in the first 100); Map B needs 3, 7, 15, 24, and 31, and its first component carries 10.9%. Map A's 2-D axes are illumination: PC1 is a brightness axis ($r = 0.986$ with mean image brightness) and PC2 a lateral gradient ($r = 0.944$); the lobes are brightness groups, not identities.

The crucial check is the rank correlation between 2-D and full-dimensional pairwise distances: Map A's plot preserves more of its space's geometry ($\rho = 0.807$) than Map B's plot preserves of the embedding space ($\rho = 0.517$). The embedding is the better representation; its 2-D picture is the less faithful one. Figure 7 shows the eigenfaces and variance curves. The projections are stable across seeds 0–2, while full-dimensional metrics are deterministic.



12 scans = six conditions × two repeats; **10 frames** per scan; one subject; captured 6 September 2026. Room light was only available from the front and left, so the design is unbalanced by necessity. Schematic only: left/right use a viewer/camera-frame convention and actual lamp positions were not recorded; front is drawn on the camera/phone side, bottom under-chin.

Each condition contributes a $(10, 512)$ matrix of unit-norm embeddings. Two repeats per condition provide the between-scan reference; frames within a scan are consecutive webcam frames, not independent photographs.

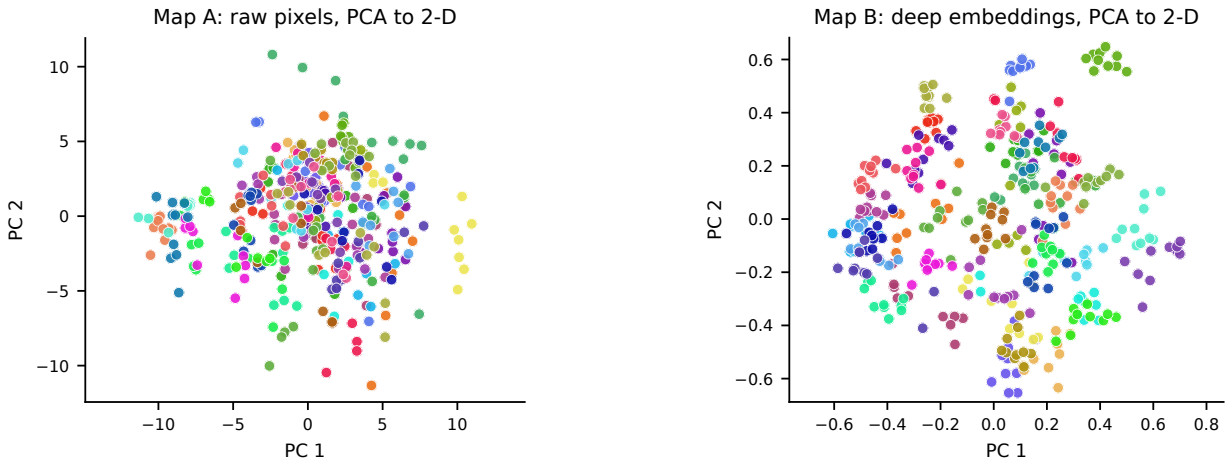
Figure 5: Webcam capture conditions. Room light and phone flash were used from up to four directions; room light was only available from front and left, so the design is unbalanced by necessity. Each captured condition has two repeat scans of ten consecutive frames; early frames are affected by webcam warm-up. The left/right placements are a schematic convention (viewer/camera frame) and do not reconstruct the actual lamp positions from the capture; this records the design, not empirically verified geometry.

6.2 Quantitative comparison

Table 1 gives the full-dimensional comparison; every row favours the embedding. Silhouette more than quadruples, 1-NN accuracy reaches 100% from 94.5%, pair ROC-AUC rises from 0.881 to 0.9995, pair EER falls from 20.5% to 1.11%, and d' roughly triples. Map A's genuine distances (8.38 ± 2.56) overlap heavily with its impostors (12.41 ± 2.45), while Map B's are nearly disjoint (0.53 ± 0.19 versus 1.33 ± 0.13; Figure 8). A rank-control projection to Map A's intrinsic span (399 components, 100% variance retained) reproduces Map A's metrics exactly (silhouette 0.10557, AUC 0.8807, EER 0.2050; maximum distance difference 3.5×10^{-12}). This shows that Map A's nominal 4096-dimensionality is not itself the source of its metrics, but it does not match the two spaces (399 versus 512) or control their intrinsic geometry; the check alone cannot tell whether dimensionality contributes to the raw-versus-embedding gap, which is measured rather than mechanistically attributed.

6.3 Why the metrics disagree

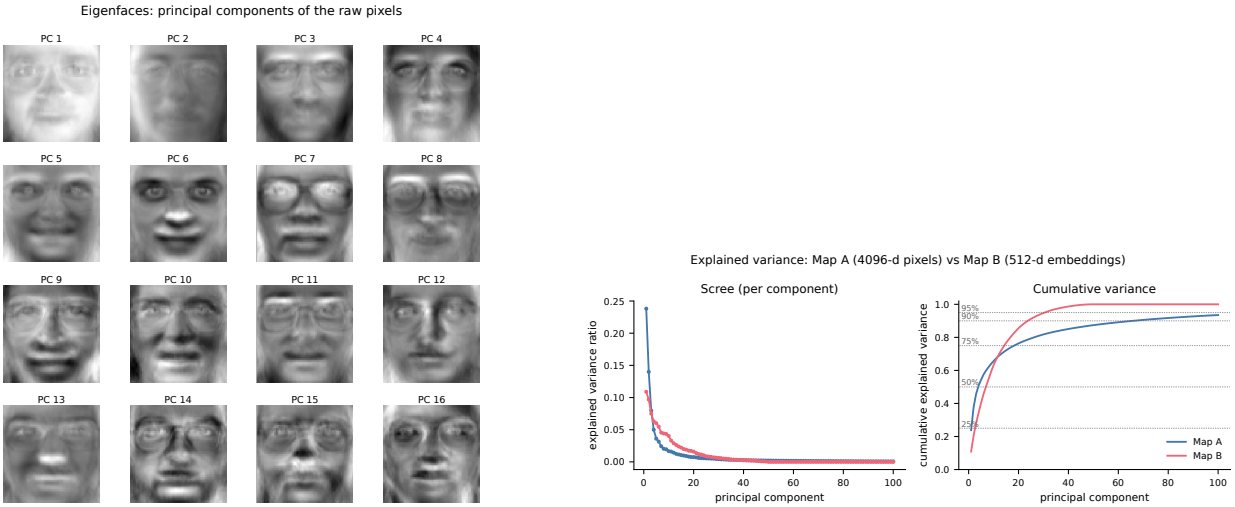
Raw pixels look respectable on identification (94.5% 1-NN) but poor on silhouette (0.106 versus 0.497) and worse on known-population verification (EER 8.7% versus 0.22%). Identification asks whether the single nearest gallery image is correct, and with ten images per identity and pre-aligned data it often is. Verification fixes one threshold and requires the distributions to be separated, not merely ordered; Map A's overlap makes that impossible, Map B's near-disjoint distributions make it easy. A good nearest-neighbour result is not evidence that a usable deployment threshold exists. The maps agree on 378 of 400 predictions (94.5%), Map B makes no errors, and there is no "both wrong" set (reported as `null`, not zero). Figure 9 summarises the metric gaps and per-identity accuracy.



(a) Map A: raw pixels.

(b) Map B: frozen embeddings.

Figure 6: The two 2-D maps with identical colour, axis, and PCA conventions. Presentation only: every metric is computed on the full 4096-d and 512-d vectors. The raw-pixel map preserves more pairwise geometry (rank correlation 0.807) than the embedding map (0.517), even though the embedding is the better representation.



(a) First 16 Eigenfaces.

(b) Explained and cumulative variance.

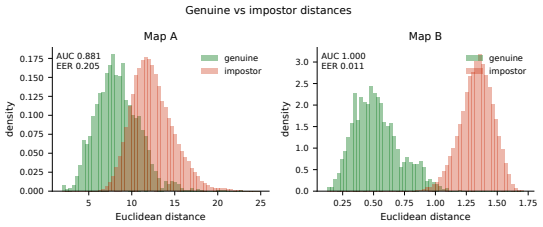
Figure 7: Map A's leading components are illumination and low-frequency shape patterns, not identity features; component 0 alone carries 23.8% of the raw-pixel variance. Map B's variance is spread over more components (31 for 95%) than Map A's concentrated variance.

6.4 Known-population verification and the threshold dial

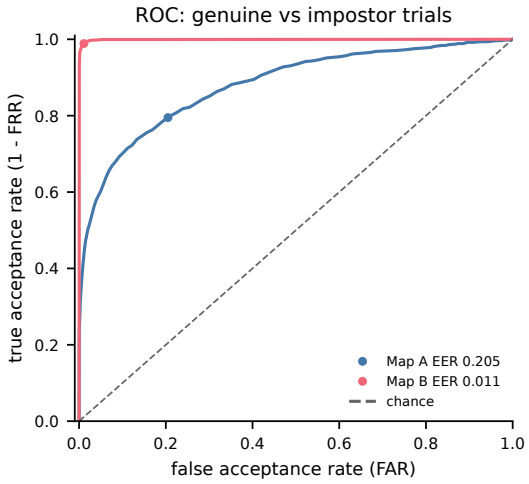
Table 2 reports the known-population (closed-set) verification protocol; all 40 identities are enrolled and there are no unknown-identity probes. Thresholds and error rates are computed from the same split scores (same-sample, exploratory), and EER is interpolated from the empirical score grid rather than an exactly realizable threshold. Map A's EER is 8.71% (sd 1.25 points) and Map B's is 0.22% (sd 0.24 points). Map B's figure is at or below the protocol's 0.5% FRR resolution (200 genuine trials per split), so read it as "at the measurement floor"; the roughly 39-fold EER ratio

Table 1: Main comparison in full dimensionality. Pair-level metrics use all 1,800 genuine and 78,000 impostor pairs from the 400 Olivetti images. The 399-component rank control reproduces every Map A row to numerical precision.

Metric	Map A: raw pixels (4096-d)	Map B: embedding (512-d)
Silhouette (40 identities)	0.1056	**0.4970**
1-NN leave-one-out accuracy	0.9450	**1.0000**
1-NN leave-one-out top-5	0.9825	**1.0000**
Pair ROC-AUC	0.8807	**0.99954**
Pair EER	0.2050	**0.01111**
TAR at FAR $= 10^{-3}$	0.2639	**0.9550**
TAR at FAR $= 10^{-2}$	0.4233	**0.9883**
d'	1.612	**4.988**
Genuine distance (mean $\pm$ sd)	8.378 ± 2.556	0.5284 ± 0.1860
Impostor distance (mean $\pm$ sd)	12.414 ± 2.450	1.3336 ± 0.1324
2-D rank correlation	**0.8070**	0.5174



(a) Genuine and impostor distances.



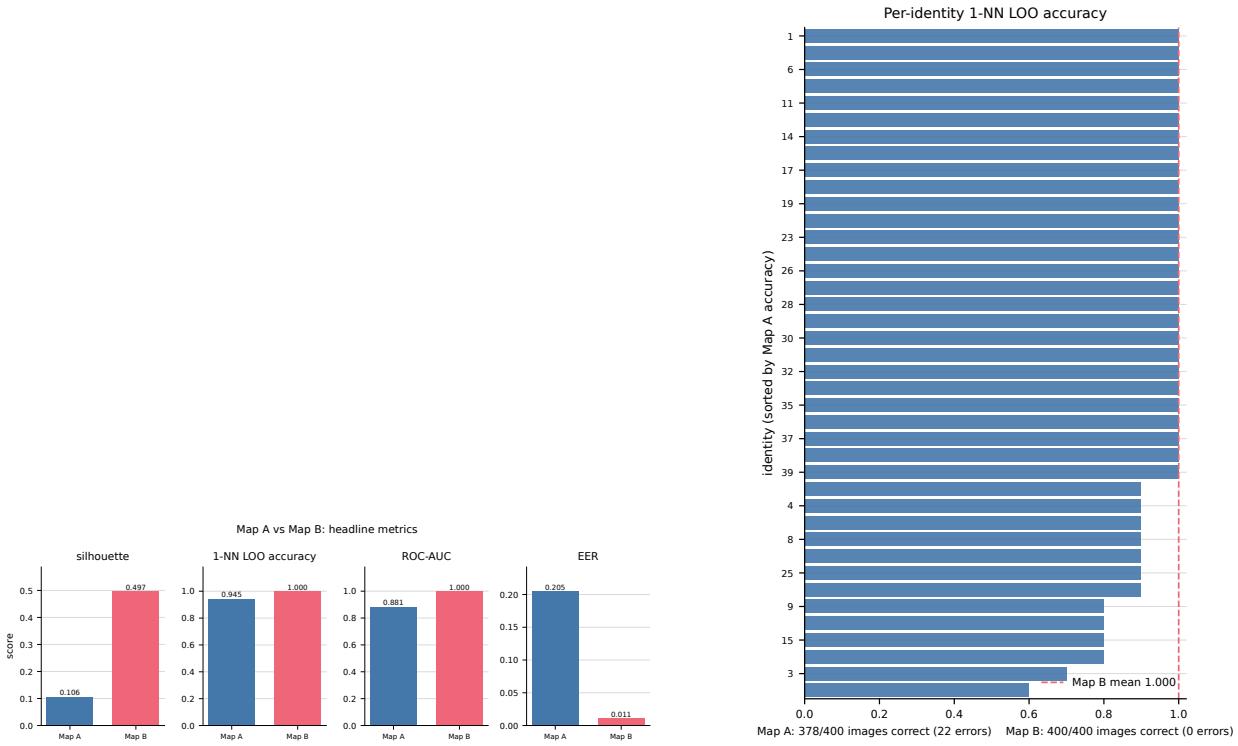
(b) Pair-level ROC with EER.

Figure 8: Pair-level verification. Map A's distributions overlap substantially; Map B's are nearly disjoint. Curves use the same 1,800 genuine and 78,000 impostor distances, not the 2-D maps.

inherits that floor. The matched-FRR comparison is not floor-limited (15,600 impostor trials per split, FAR resolution 0.0064%): at 1% false-reject rate, Map A accepts 48.59% of impostors and Map B 0.044%, so Map A admits roughly 1,115 times as many impostors at the same convenience. At a 1% false-accept rate, Map B accepts 99.9% of genuine probes versus 74.4% for Map A.

Figure 10 shows the curves on separate axes: the EER thresholds are 8.505 and 0.8396, roughly ten times apart, and a shared axis would invite a false comparison of threshold values. Within the sampled thresholds Map A has no threshold with both error rates small, while Map B has a broad operating region. Robustness checks do not change the ranking: the fixed first-half/second-half split gives EERs 9.00% and 0.256%, the mean-template rule 11.85% and 0.22%. Pair-level EERs (20.5% and 1.11%) come from a different, pair-based protocol and agree on ranking but not absolute value; the known-population verification protocol supplies the headline numbers.

An additive identity-disjoint check separates threshold selection from evaluation. Thirteen identities calibrate the threshold, thirteen identity-disjoint identities form the evaluation gallery, and fourteen identities are unknown to that gallery; across five reused partitions, thresholds are chosen



(a) Metric summary.

(b) Per-identity 1-NN accuracy.

Figure 9: Metric summary and per-identity accuracy. The embedding wins every metric; Map B is perfect on every identity, while Map A's errors concentrate in a few identities. Cross-map prediction agreement is 94.5%.

on calibration impostor scores only. At a nominal 1% calibration FAR, Map A attains held-out TAR 75.08% with held-out FAR 1.21% and accepts 18.86% of unknown-to-gallery probes; Map B attains held-out TAR 100% with held-out FAR 0.77% and accepts 12.14% of unknown probes. The ranking survives identity-disjoint calibration, but the unknown-identity problem remains large: a never-enrolled Olivetti identity is still accepted as somebody in 12.1% of Map B trials. The 12.14% unknown-accept figure is an any-of-13-gallery-claims rate (an unknown probe is accepted if any of the 13 evaluation identities accepts it), whereas the 0.77% figure is the per-claim held-out FAR; they use different denominators and are not contradictory. The 1% calibration target is not recovered exactly out of sample, and the five partitions reuse the same identities and are not independent samples.

6.5 Webcam illumination study

Table 3 summarises the study. The frame-to-frame noise floor is 0.186 (per-scan means 0.114–0.334), and same-condition repeats differ by 0.227–0.371. The front-lit room-versus-flash comparison yields mean distance 0.258 (range 0.168–0.384), inside the repeat-scan range. The six flash-direction comparisons average 0.714, 3.83 times within-scan jitter. These sequential captures also vary unmeasured pose and auto-exposure, so the differences cannot be attributed to brightness or light direction alone. Front-to-side pairs are large (0.636–0.733) while left–right is smaller (0.511); we read this as a hypothesis that two half-shadowed faces resemble each other more than either resembles a front-lit face, not as an established property of the embedding. Side-to-bottom is worst (0.808–0.888);

Table 2: Known-population (closed-set) verification: five random 5/5 splits (seeds 0–4), 200 genuine and 15,600 impostor trials per split. Every score is the minimum distance from a query to the five enrolment images of the claimed identity; accept if score $\leq \tau$. All identities are known and enrolled, so FAR is cross-identity error, not open-set stranger rejection. Mean (sd) across splits; thresholds are selected on the same scores used to report errors (exploratory), and EER is interpolated. Map B's EER is at the protocol's 0.5% FRR resolution.

Operating point	Map A	Map B
EER	0.0871 (0.0125)	0.00224 (0.00244)
EER threshold	8.505	0.8396
TAR at FAR $= 10^{-3}$	0.590	0.994
TAR at FAR $= 10^{-2}$	0.744	0.999
FAR at matched FRR = 1%	0.4859 (0.1189)	0.000436 (0.000378)
AUC	0.96999	0.99997

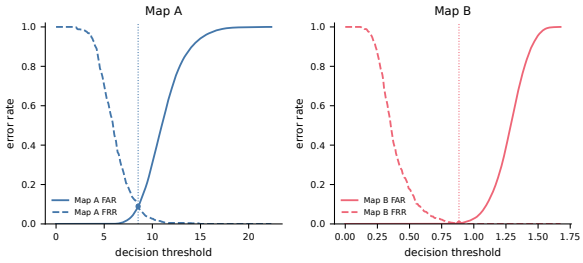


Figure 10: FAR (solid) and FRR (dashed) against threshold, with EER marked. Separate axes are used because the representations produce distances on different scales (EER thresholds 8.505 versus 0.8396). Map A has no threshold with both error rates small; Map B has a wide operating region.

this is an observation for one subject, and a training-data explanation would be speculation. Pooling frames reduces mean distances by 4.82% (0.604 to 0.575), a modest gain that does not change condition ordering (Figure 11).

The worst same-person pair is flash/right repeat 2 against flash/bottom repeat 2: 0.967 frame-level (cosine similarity 0.532, cosine distance 0.468), 0.915 after scan averaging (cosine similarity 0.581, cosine distance 0.419). In this observed worst pair, a frame-level threshold below 0.967 would have rejected the enrolled sample (below 0.915 for scan means). This describes the captured pairs only: it is not a validated lower bound for deployment, because unobserved conditions could be worse and the sample is one person. That is the convenience side of the dial, measured on one person. Frames are not independent (mean within-scan distance grows from 0.153 at lag 1 to 0.257 at lag 9), so the effective sample size per scan is below ten; frame 0 is also a warm-up outlier (0.243 versus 0.167 at frame 5).

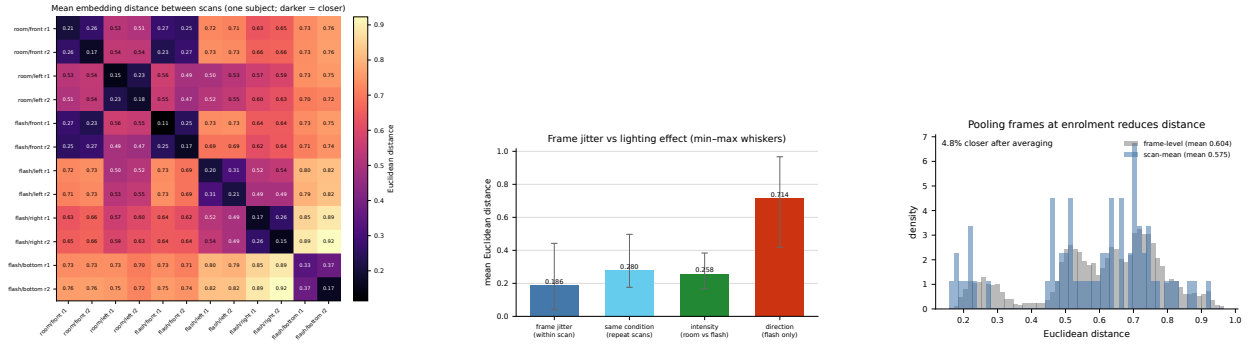
The smallest distance from any Olivetti embedding to any live scan is 0.887 (frame-level 0.880; mean of per-Olivetti minima 1.288). It is domain-confounded—different capture device, crop, resolution, and people—so it is reported only as a descriptive cross-domain distance, not as a calibrated impostor distribution and neither as an upper nor a lower bound on webcam-domain stranger distance (Section 7).

7 Discussion

What the comparison shows. The embedding dominates every identity metric, but the interpretation is narrower than "deep beats classical". Map A's reported metrics use original full-dimensional

Table 3: Webcam illumination study summary (Euclidean distances between unit-norm embeddings; cosine similarity $= 1 - d^2/2$). The noise floor is the mean within-scan frame-pair distance over twelve scans. The worst genuine pair is flash/right repeat 2 versus flash/bottom repeat 2. The cross-domain row is domain-confounded and is reported only as a descriptive distance, not as a calibrated impostor distribution and neither as an upper nor a lower bound on webcam-domain stranger distance.

Quantity	Value
Within-scan jitter (noise floor)	0.186
Same-condition between-scan means	0.227–0.371
Room/front vs flash/front (observed)	0.258
Six flash-direction pairs (observed mean)	0.714
Flash-pair / within-scan jitter ratio	3.83
Pooling reduction (scan means vs frames)	4.82%
Worst genuine frame pair	0.967
Worst genuine after scan averaging	0.915
Cross-domain minimum (Olivetti to live scan)	0.887



(a) 12×12 scan distances.

(b) Within-scan and condition contrasts.

(c) Frame-level vs scan means.

Figure 11: Observational live study of one subject. Front-lit room-versus-flash distances fall within the same-condition repeat range, while the mean of the flash-direction pair distances is 3.83 times within-scan jitter. Separate scans and uncontrolled pose/exposure preclude a causal lighting effect. Pooling ten frames modestly reduces observed distances.

pixels; its fitted PCA basis is for the plots and full-rank isometry check, not for verification. Map B is a 23.5-million-parameter network trained on 3.31 million images and then frozen. The gap therefore combines representation, training corpus, and architecture, which this study cannot separate. Fitting PCA on the probe images makes the visualisations transductive but cannot establish that the full-vector verification gap is conservative.

The metrics disagree, and that is the point. Raw pixels look respectable on identification (94.5% 1-NN) but poor on silhouette and substantially worse on known-population verification. Identification asks whether the single nearest gallery image is correct, and with ten images per identity and pre-aligned data the nearest genuine image is often close enough. Verification fixes one threshold and requires the genuine and impostor distributions to be separated, not merely ordered. Map A's overlap makes that impossible; Map B's near-disjoint distributions make it easy. A good nearest-neighbour result is not evidence that a usable deployment threshold exists.

The prettier map is the worse space. Map A's 2-D plot is a more faithful projection of its own space ($\rho = 0.807$) than Map B's plot is of the embedding space ($\rho = 0.517$), and Map A's 2-D

axes are literally brightness and a lateral lighting gradient. A viewer comparing the two scatters is not comparing recognition ability: the plots preserve unequal fractions of the geometry they summarise. Raw-pixel variance is dominated by a few illumination directions, so two components capture much of its structure even though that structure is not identity; the embedding's leading variance directions are less identity-discriminative, so compressing it discards more of what makes it useful.

Observed lighting-condition contrasts. The two front-lit room-versus-flash groups differ by mean distance 0.258, within the 0.227–0.371 same-condition repeat range. Across four flash-direction labels the mean pair distance is 0.714, or 3.83 times within-scan jitter, with side-to-bottom pairs largest. These are separate sequential captures of one subject: pose and auto-exposure are unmeasured, so the study does not identify a causal intensity or direction effect. Left–right (0.511) is smaller than front–side pairs, consistent with a tentative "half-shadowed" region. Adding a side-lit enrolment scan is a hypothesis for a future held-out verification experiment, not a validated policy.

The security–convenience dial. At matched 1% FRR, Map A accepts 48.6% of impostors and Map B 0.044%; at 1% FAR, Map B accepts 99.9% of genuine probes against 74.4% for Map A. Within the sampled operating points Map B has a broad region with both error rates small, while Map A has none. Thresholds do not transfer: the verification EER thresholds are 8.51 for pixels and 0.84 for embeddings (the pair-level EER thresholds are different again), and webcam distances live on a third scale. The cross-domain check makes the point: the smallest Olivetti-to-webcam distance (0.887) is below the worst same-person webcam distance (0.967), but the two numbers are not comparable scores, because the cross-domain value is confounded by device, crop, resolution, and population and is not a calibrated impostor score. It is not evidence that a stranger would be accepted, and it is neither an upper nor a lower bound on webcam-domain stranger distance. Domain shift is large enough to make a threshold fitted on one dataset meaningless on another; any deployment must calibrate on in-domain impostors, which this study cannot provide because it has no stranger capture. The same-sample operating points in Section 6 are therefore exploratory, and an additive identity-disjoint check does not close the gap: at a nominal 1% calibration FAR, Map B reaches held-out TAR 100% against 75.08% for Map A, but it still accepts 12.1% of never-enrolled identities as somebody (any-of-13-gallery claims; Map A 18.9%), and the nominal 1% FAR is not validated out of sample (per-claim held-out FAR is 0.77%). Unknown-identity rejection remains the open problem.

8 Limitations

One subject, no stranger. The webcam study is observational: one subject, conditions fixed by the capture session rather than randomised, and no stranger was ever recorded. Every live distance is therefore same-person and measures convenience, not security. No live threshold can be calibrated from these data, because there is no impostor distribution in the live domain. The cross-domain Olivetti-to-webcam distances are descriptive only; they are not a calibrated impostor distribution and neither an upper nor a lower bound on webcam-domain stranger distance.

Known-population impostors. The Olivetti verification protocol is closed-set. Its impostors are the other *enrolled* identities from the same 40-person dataset, captured in the same session, so the reported false-accept rates measure cross-identity confusion within a known population. The primary analysis contains no unknown-to-gallery probes and no stranger trials, so its numbers cannot be read as open-set rejection of unknown people or as deployment false-accept rates. An

additive identity-disjoint check (13 calibration, 13 evaluation, and 14 unknown-to-gallery identities; five reused partitions; 65 genuine, 1,560 impostor, and 140 unknown probes per partition) selects thresholds on calibration impostor scores only. At a nominal 1% calibration FAR it reaches held-out TAR 75.08% (Map A) and 100% (Map B), but held-out FAR is 1.21% and 0.77% respectively, so the 1% target is not validated out of sample; 12.1% of unknown-to-gallery identities are still accepted as somebody by Map B (any-of-13-gallery-claims rate; 18.9% for Map A), while the per-claim held-out FAR is 0.77%; the two use different denominators. Even the unknown-to-gallery identities remain known to the frozen model and come from the same dataset, and the partitions overlap rather than providing independent samples.

Threshold selection and interpolated EER. The thresholds at matched FRR and at the EER are selected on the same split scores used to report false-accept and false-reject rates. This is in-sample, descriptive calibration, not a held-out calibration procedure, and it makes the reported operating points optimistic. The EER is interpolated from a finite score grid (FRR resolution 0.5%, FAR resolution 0.0064%) and is not necessarily an exactly realizable distance threshold; empirical FAR and FRR at explicit thresholds should be preferred when a specific operating point matters. Map B's 0.22% EER is below the FRR grid resolution and should be read as "at the measurement floor", not as three significant figures.

Small, correlated samples. There are 400 images, 40 identities, and ten images per identity, all from a single capture session. The 1,800 genuine pairs are not independent (each image appears in nine pairs), and the five 5/5 splits reuse the same images, so split variance is not capture variance. Image-level resampling does not model identity-level or session-level dependence, and no confidence interval here accounts for it; the effective sample size is much smaller than the nominal trial counts.

Olivetti is the best case for raw pixels. The dataset is tightly cropped and roughly eye-aligned, removing the geometric variability that most damages pixel-space methods. Map A's results are an optimistic reading for raw pixels, and the same property makes the deep model's task easy, so both maps' absolute accuracy is optimistic relative to unconstrained data.

Transductive visualisation and frozen model. PCA for both 2-D plots is fit on all 400 images, including verification probes. Recognition metrics use the unprojected vectors, so this fit does not show that the measured advantage of Map B is conservative. The embedder was not trained or fine-tuned on either evaluation dataset, but it was pretrained on millions of faces; the observed gap combines architecture, training data and representation and does not isolate any one cause.

The rank control does not explain the gap. Projecting Map A to its intrinsic 399-dimensional span reproduces its metrics because the projection is distance-preserving, but 399 is not 512 and the two representations have different intrinsic geometry. The control rules out nominal input dimensionality as the source of Map A's metrics; it does not provide a matched-dimensionality explanation of the raw-versus-embedding gap, and no such explanation is claimed.

Webcam frames are not independent. The ten frames per scan are consecutive webcam frames of a moving person: within-scan distance grows from 0.153 at lag 1 to 0.257 at lag 9, and the first frame is a warm-up outlier. Frame-pair counts overstate the effective sample size, which is why scan means are the more meaningful unit. Raw webcam frames were not retained, so alternate crops, detectors, or preprocessing cannot be re-analysed from the saved data.

Illumination comparison is observational. The intensity comparison contrasts room light with phone flash, so it changes source spectrum, source size, and shadow structure as well as brightness, not brightness alone. The direction comparison changes the phone position but remains one subject, one session, and two repeats per condition. Room-front and flash-front were captured in separate scans, so no causal attribution to brightness is made. The "half-shadowed region" reading is a hypothesis, and the under-lighting observation is descriptive; neither is a controlled illumination experiment.

Detection and thresholds. The MTCNN implementation crops the O-Net box without a landmark similarity transform, so alignment is detector-box alignment; a landmark-warped pipeline might change the between-condition distances, which this study does not test. Olivetti verification thresholds (EER thresholds 8.51 and 0.84) and the webcam distance scale are not comparable, and no threshold in this paper should be deployed.

No demographic, sensor, or session diversity. One dataset with limited demographic variety, one laptop webcam, one model and backbone, and no repeated sessions across days. The study cannot speak to fairness, sensor generalisation, or template ageing. The optional synthetic-lighting ablation was not run.

9 Reproducibility

All experiments ran on an Apple M5 Pro (18 cores, 48 GB unified memory) under macOS 27.2 (arm64), Python 3.12.14, NumPy 2.5.3, scikit-learn 1.9.1, PyTorch 2.14.0, `facenet-pytorch` 2.5.3, and Matplotlib 3.11.2. The embedder runs on Apple MPS; MTCNN runs on CPU because its operations are unsupported by MPS. No training runs or cloud GPUs were used. Every experiment uses seed 0 unless a protocol averages seeds 0–4, and every sklearn estimator receives `random_state=0`. The CNN is deterministic in evaluation mode, and its Olivetti embeddings are cached. Results are written to JSON by the analysis scripts. The numerical tables and prose were transcribed by hand from those files into LaTeX, not generated from JSON; they must be reviewed against the JSON whenever results are regenerated. Each results file records the git commit, Python and package versions, device, and seed.

From the source checkout, a portable entry point is:

```
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements.txt
python -m pip install -e . --no-deps
make all          # tests, maps, live, verification, architecture
make diagrams     # TikZ figure sources -> vector PDFs
make paper        # two LaTeX passes
```

The exact experiment environment and cached model checkpoint are not bundled with an installed wheel; installed users may set `FEM_PROJECT_ROOT` to a source checkout for data paths. The initial fetch of Olivetti faces or model weights may require network access.

The checkpoint is `20180402-114759-vggface2.pt`, cached locally (111,898,327 bytes; SHA-256 begins `281cebca8662831a987a874bdcb36e73`). Its measured layer shapes and parameter counts are in `results/model_arch.json`; Figure 3 is manually drawn from those measurements and must be checked against them after changes. Olivetti Faces is fetched through scikit-learn's loader and cached locally; the dataset is not bundled with this repository. The results JSON files, scan manifest

and figure-generation code are in the project repository. The webcam embeddings are biometric templates of an identifiable person and are not distributed, so the webcam study can be checked against `results/live.json` but not re-run from a fresh clone; raw webcam frames were not retained. The analysis code is a library (`fem/`) plus experiment runners (`analysis/`), so the comparison re-runs without duplicating preprocessing or metrics. Small numerical differences between BLAS implementations are possible.

10 Future work

Yale B. Extended Yale Face Database B (Georghiades et al., 2001) sweeps lighting around the room for roughly 38 people at fixed pose. Re-running the two-map comparison there would test the prediction this paper can only sketch—raw pixels should cluster by lamp position, the embedding should remain identity-driven—with enough subjects to support a claim about people rather than one face.

LFW and alignment. Olivetti is the best case for raw pixels, so the comparison should be repeated on unconstrained data such as LFW (Huang et al., 2007). An intermediate step is an MTCNN-augmented Olivetti ablation: Map A on detector-cropped rather than pre-aligned images, and Map B on both crops, separating alignment from representation.

Training and representations. The project plan reserved training for a final stage; it remains out of scope. Training a small embedding with triplet loss on Olivetti would show how much of the gap is available from the data at hand without a large pretrained corpus, and would require held-out identities. Repeating it with ArcFace (Deng et al., 2019) would test whether the conclusions are properties of deep embeddings generally or of this checkpoint.

More subjects, strangers, sessions. Several subjects, stranger images for impostor calibration, repeated sessions across days, and different sensors would turn the live study into a calibration experiment and let the security–convenience dial be set with data rather than described.

Synthetic relighting. Applying smooth illumination gradients to Olivetti images and re-embedding them would isolate direction from capture artefacts and connect the webcam observations to the illumination-cone literature.

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